

EXPERIMENT 6

Write a Python Program for the Vacuum Cleaner Problem

Aim

To develop a Python program that simulates the **Vacuum Cleaner Problem** using a **Simple Reflex Agent** to clean all dirty rooms and demonstrate intelligent agent behavior in Artificial Intelligence.

Objective

- To understand the concept of **Simple Reflex Agents**.
- To implement the Vacuum Cleaner Problem using Python.
- To simulate an intelligent agent that senses and acts according to the environment.
- To study how AI agents make decisions based on percepts.

Theory

The **Vacuum Cleaner Problem** is one of the simplest examples of an **Artificial Intelligence Intelligent Agent**. The environment consists of two rooms, **Room A** and **Room B**, each of which may be **Clean** or **Dirty**. The vacuum cleaner agent can perform the following actions:

- Move Left
- Move Right
- Suck (Clean)
- No Operation (Stop)

The agent observes its current location and the cleanliness of that room. If the room is dirty, it cleans it. Otherwise, it moves to the other room. Once both rooms are clean, the agent stops.

This problem demonstrates how an intelligent agent perceives its environment through sensors, makes decisions, and performs actions using actuators.

Algorithm

Step 1: Initialize the environment with two rooms (Room A and Room B).

Step 2: Assign each room a status (**Clean** or **Dirty**).

Step 3: Place the vacuum cleaner in one of the rooms.

Step 4: Sense the status of the current room.

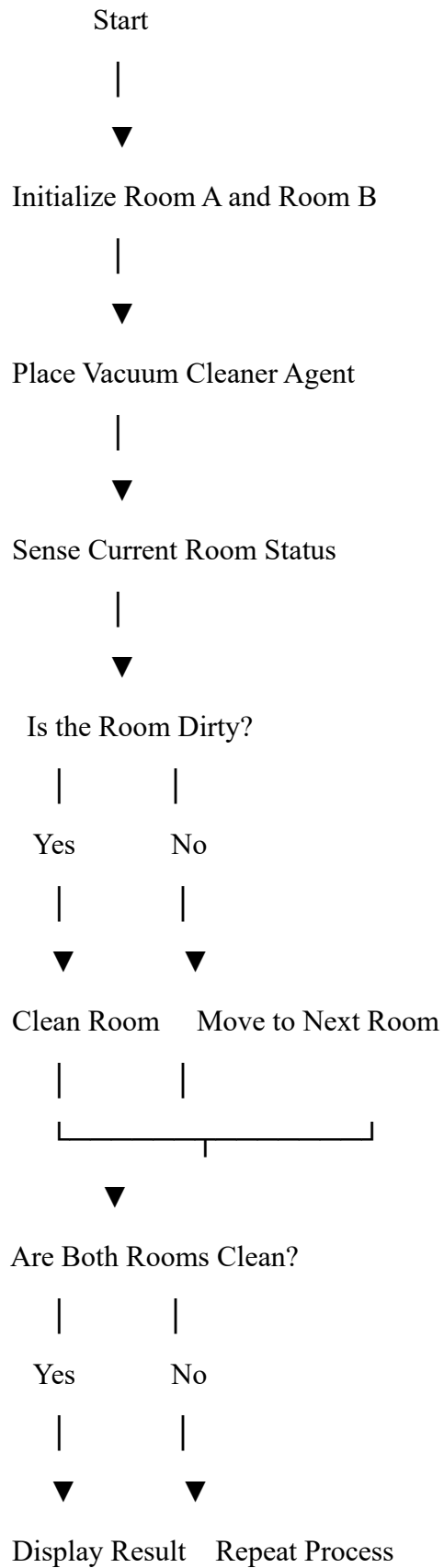
Step 5: If the room is **Dirty**, perform the **Suck** action and clean the room.

Step 6: If the room is **Clean**, move to the other room.

Step 7: Repeat Steps 4 to 6 until both rooms become clean.

Step 8: Display the final status of both rooms.

Flowchart



|



Stop

Python Program

```
def vacuum_cleaner(room_status, location):
```

```
    while True:
```

```
        if room_status[location] == "Dirty":
```

```
            print("Cleaning Room", location)
```

```
            room_status[location] = "Clean"
```

```
        if location == "A":
```

```
            location = "B"
```

```
        else:
```

```
            location = "A"
```

```
        if room_status["A"] == "Clean" and room_status["B"] == "Clean":
```

```
            break
```

```
    print("\nFinal Room Status")
```

```
    print(room_status)
```

```
rooms = {
```

```
    "A": "Dirty",
```

```
    "B": "Dirty"
```

```
}
```

```
vacuum_cleaner(rooms, "A")
```

Sample Output

Cleaning Room A

Cleaning Room B

Final Room Status

{'A': 'Clean', 'B': 'Clean'}

Explanation

Initially, both **Room A** and **Room B** are dirty.

The vacuum cleaner starts in **Room A**, senses that it is dirty, and cleans it.

The agent then moves to **Room B**, detects that it is dirty, and cleans it.

After both rooms become clean, the agent stops execution.

This demonstrates how a **Simple Reflex Agent** performs actions based solely on the current percept.

Advantages

- Easy to understand and implement.
- Demonstrates the working of a **Simple Reflex Agent**.
- Suitable for simple and fully observable environments.
- Requires very little memory.

Applications

- Robotic vacuum cleaners.
- Home automation systems.
- Industrial cleaning robots.
- Intelligent household appliances.
- Basic AI agent simulations.

Result

The Python program successfully simulated the **Vacuum Cleaner Problem** using a **Simple Reflex Agent**. The agent sensed the environment, cleaned all dirty rooms, and stopped after completing the task.

Conclusion

The **Vacuum Cleaner Problem** is a fundamental Artificial Intelligence problem used to demonstrate the behavior of a **Simple Reflex Agent**. The agent senses the current state of the environment and performs appropriate actions without maintaining memory of previous states. This experiment illustrates the concepts of intelligent agents, perception, action, and rational behavior in AI. It provides a foundation for understanding more advanced agent architectures such as **Model-Based**, **Goal-Based**, and **Utility-Based Agents**.

```
PS C:\Users\Varshini M\AppData\Local\Programs\Microsoft VS Code> & "C:\Python39\python.exe" "c:/Users/Varshini M/OneDrive/Desktop/Here!!!/ARTIFICIAL INTELLIGENCE/EXP1.py"
Cleaning Room A
Cleaning Room B

Final Room Status
{'A': 'Clean', 'B': 'Clean'}
PS C:\Users\Varshini M\AppData\Local\Programs\Microsoft VS Code>
```

```
C: > Users > Varshini M > OneDrive > Desktop > Here!!! > ARTIFICIAL INTELLIGENCE > VSC EXPs > EXP1.py > ...
1  def vacuum_cleaner(room_status, location):
2
3      while True:
4
5          if room_status[location] == "Dirty":
6              print("Cleaning Room", location)
7              room_status[location] = "Clean"
8
9          if location == "A":
10             location = "B"
11         else:
12             location = "A"
13
14         if room_status["A"] == "Clean" and room_status["B"] == "Clean":
15             break
16
17         print("\nFinal Room Status")
18         print(room_status)
19
20     rooms = {
21         "A": "Dirty",
22         "B": "Dirty"
23     }
24
25     vacuum_cleaner(rooms, "A")
26
```